UNITED STATES INTERNATIONAL UNIVERSITY

Advanced E-Commerce Analytics:

The Olist Marketplace Case Study

Business Intelligence and Analytics
Power BI End Semester Examination — DSA 3050A

E-Commerce Analytics	Data Modeling	DAX Intelligence	Dashboard Design

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1. Introduction

This section addresses the domain, the business problem, and the suitability for advanced analysis.

Project Overview

This project focuses on the **E-commerce Retail Analytics** domain, specifically analyzing the marketplace operations of **Olist**, the largest department store in Brazilian marketplaces.

Business Problem Statement

As a Business Intelligence Analyst, the goal is to resolve a critical visibility gap regarding Olist's operational health. The business needs to understand:

- **Sales Performance:** Which regions and product categories drive the highest revenue?
- **Logistics Efficiency:** How does the 'Delivery Lag' (difference between estimated and actual delivery) affect customer satisfaction?
- **Profitability:** Are high freight costs in certain Brazilian states eating into the net margins?

Suitability for Advanced Power BI Analysis

The Olist dataset is uniquely suited for this examination because it offers:

- **High Volume:** The main transactional table contains over 99,000 orders, satisfying the 9,000-row requirement.
- **Rich Data Types:** It includes geographic data (States/Cities), temporal data (Timestamps), and numerical data (Price/Freight), allowing for advanced Time Intelligence and Geospatial analysis.
- **Relational Model Requirement:** The data is spread across 8 distinct files, necessitating the creation of a sophisticated Star Schema with complex relationships.

2. Dataset Description

This section addresses the source, key tables, fields, and the visual evidence of the raw data.

Data Source

The data is the **Brazilian E-Commerce Public Dataset** provided by Olist and hosted on Kaggle. It contains real, anonymized commercial data from 2016 to 2018.

Key Tables and Fields

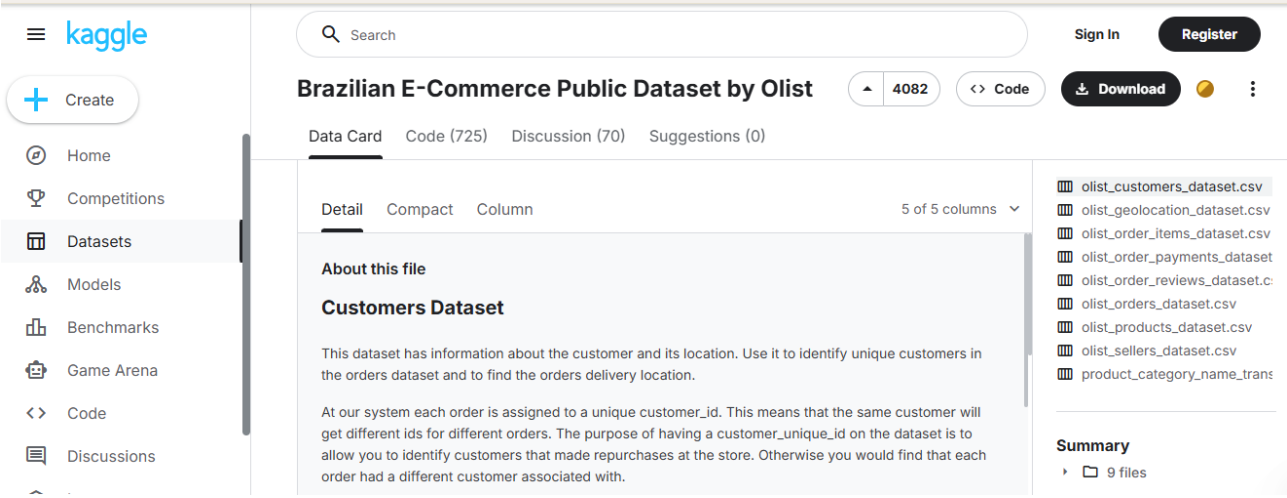

Figure 1: Brazilian E-Commerce Public Dataset on Kaggle showing all 9 source files

Table Name	Role	Key Fields
olist_orders_dataset	Fact Table	order_id, customer_id, order_status, order_purchase_timestamp
olist_order_items	Fact/Transaction	order_id, product_id, price, freight_value
olist_products	Dimension	product_id, product_category_name
olist_customers	Dimension	customer_id, customer_zip_code_prefix, customer_state
olist_order_payments	Fact/Detail	order_id, payment_type, payment_value
product_category_translation	Lookup	product_category_name, product_category_name_english

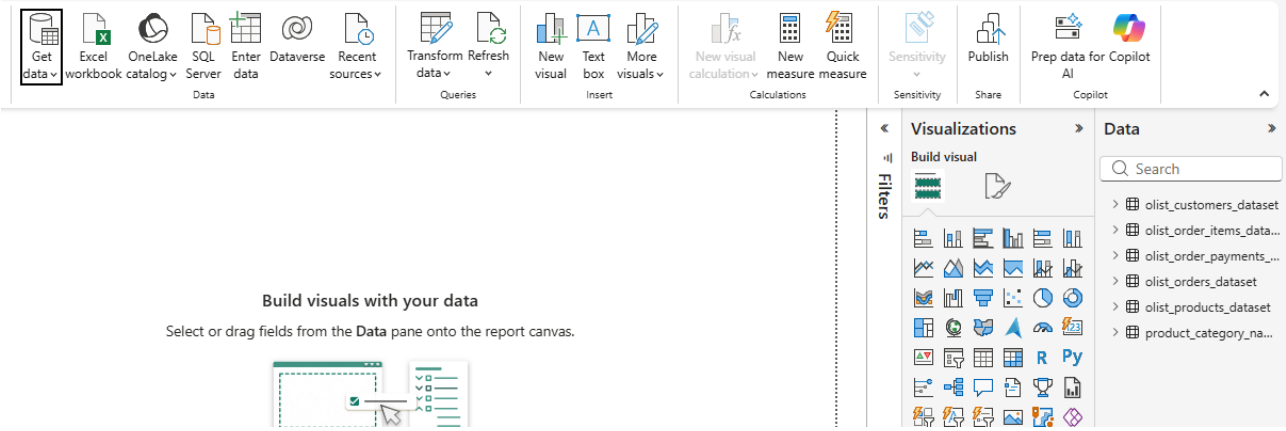

Figure 2: Raw data loaded into Power BI — all 5 tables visible in the Data pane

3. Data Cleaning and Transformation in Power Query

A total of **11 meaningful transformation steps** were performed across the dataset tables in Power Query Editor, exceeding the minimum requirement of 8. Each step is documented with technique, rationale, and before/after screenshot evidence.

1 Load the Data

What I did:	I first loaded the data to the Power BI interface in order to begin the transformation process.
Why it was necessary:	All source files must be imported before any transformations can be applied. This establishes the foundational query connections.
Technique used:	*CSV Import via Get Data > Text/CSV connector*

2 Merging Product Translations for Localization

What I did:	I used the Merge Queries feature to join the olist_products_dataset with the product_category_name_translation table using the product_category_name column as a common key.
Why it was necessary:	The original dataset contained product categories in Portuguese. To ensure the final dashboard is accessible to English-speaking stakeholders, I merged the tables to bring in the English equivalents.
Technique used:	*Merge Queries — Left Outer Join on product_category_name*

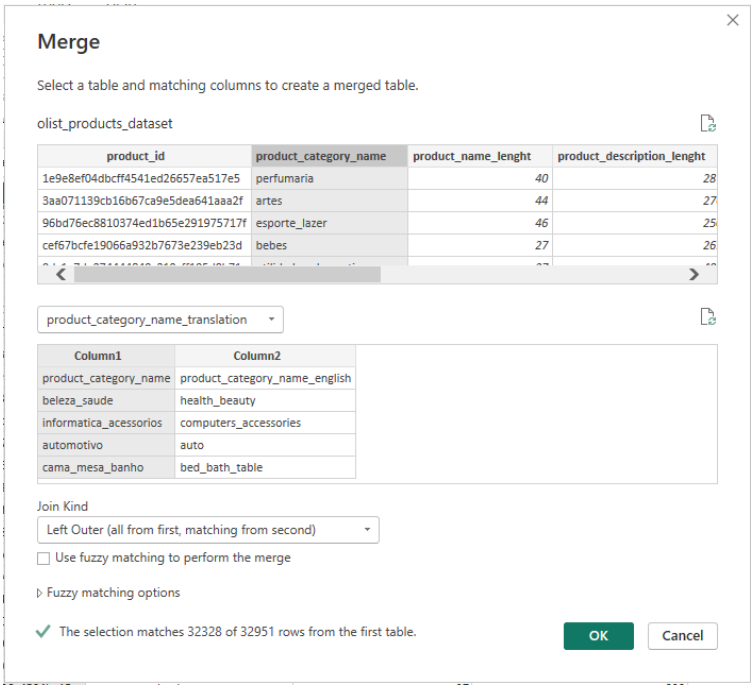

Figure 3: Merge Queries dialog joining olist_products_dataset with the translation table

Before / After — Product Category Language

Before: Portuguese categories	After: English categories added

product_category_name
perfumaria
perfumaria
artes
beleza_saude
beleza_saude
beleza_saude
esporte_lazer
esporte_lazer
esporte_lazer
esporte_lazer
esporte_lazer
esporte_lazer
esporte_lazer
informatica_acessorios
informatica_acessorios

Column2
perfumery
perfumery
art
health_beauty
health_beauty
health_beauty
sports_leisure
sports_leisure
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sports_leisure
computers_accessories
computers_accessories
computers_accessories

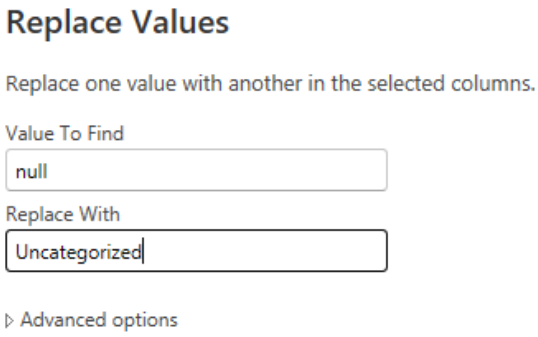


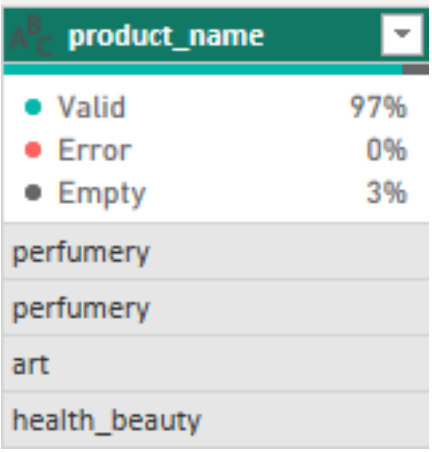
Figure 4: Expanded product table with English category name column

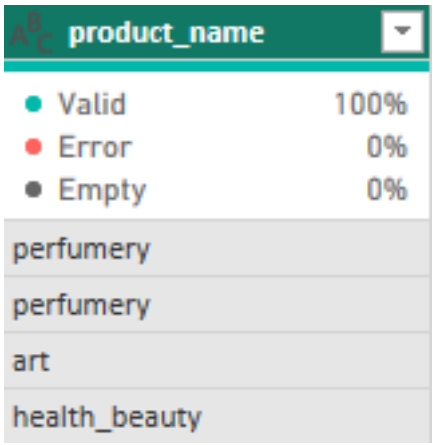
3 Handling Null Values in Product Categories

What I did:	After expanding the English category names, I identified several null values. I used the Replace Values transformation to change all null entries to 'Uncategorized'.
Why it was necessary:	To maintain data integrity. Without this, 'null' would appear as a blank space in charts, which obscures the true count of products sold and looks unprofessional in a final report.
Technique used:	*Replace Values: null → 'Uncategorized'*

Before / After — Null Value Handling

Before: 3% empty values	After: 100% valid (0% empty)





4 Consolidating Transactional Data — Master Fact Table

What I did:	I performed a multi-step merge starting with the olist_orders_dataset table, joining it with order_items and order_payments based on the order_id. I renamed the resulting master table to Fact_Sales.
Why it was necessary:	This follows the Star Schema principle. By consolidating transactional data (prices, shipping, and payments) into one table, I simplified the data model, ensuring DAX measures are more efficient and relationship logic is easier to manage.
Technique used:	*Table Merging and Master Fact Consolidation (multi-step Left Outer joins)*

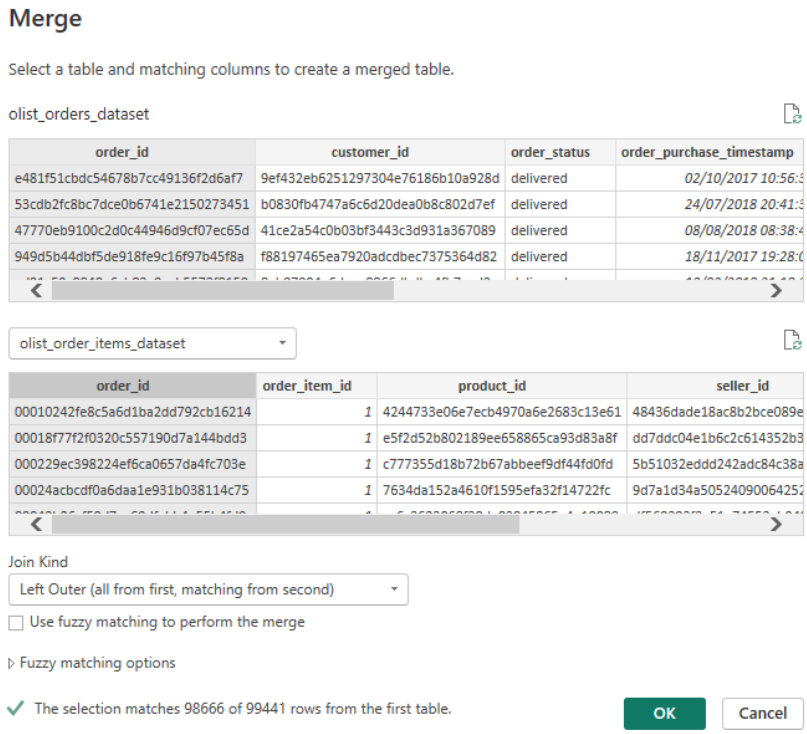


Figure 5: Merge dialog joining olist_orders_dataset with olist_order_items_dataset on order_id

Column selection when expanding merged tables:

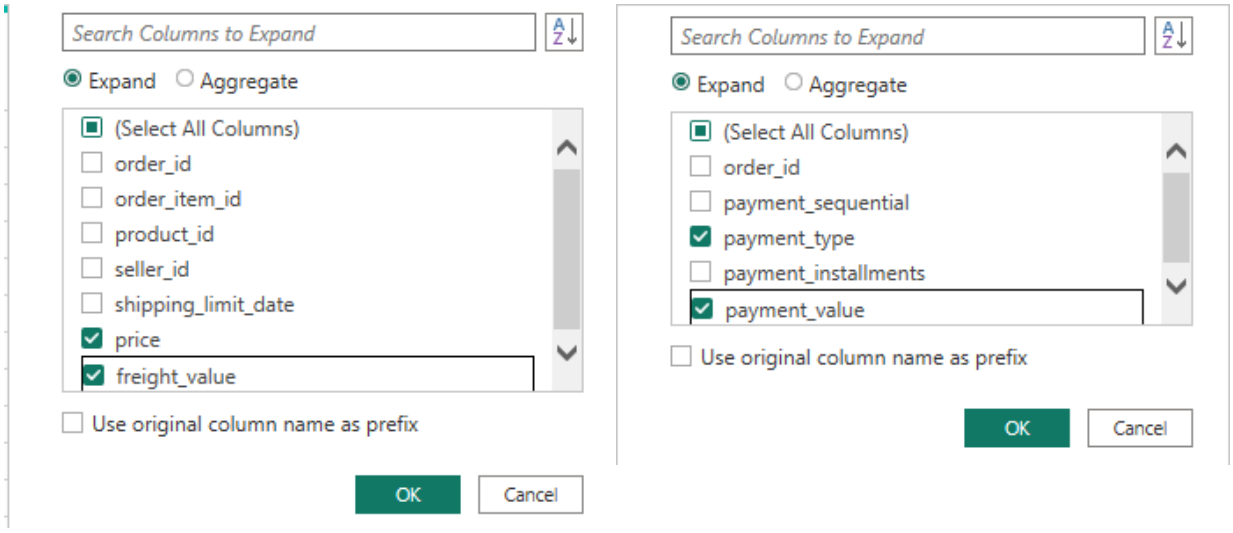


Figure 6: Left — expanding order_items columns (price, freight_value); Right — expanding order_payments columns (payment_type, payment_value)

Result of the multi-table merge:

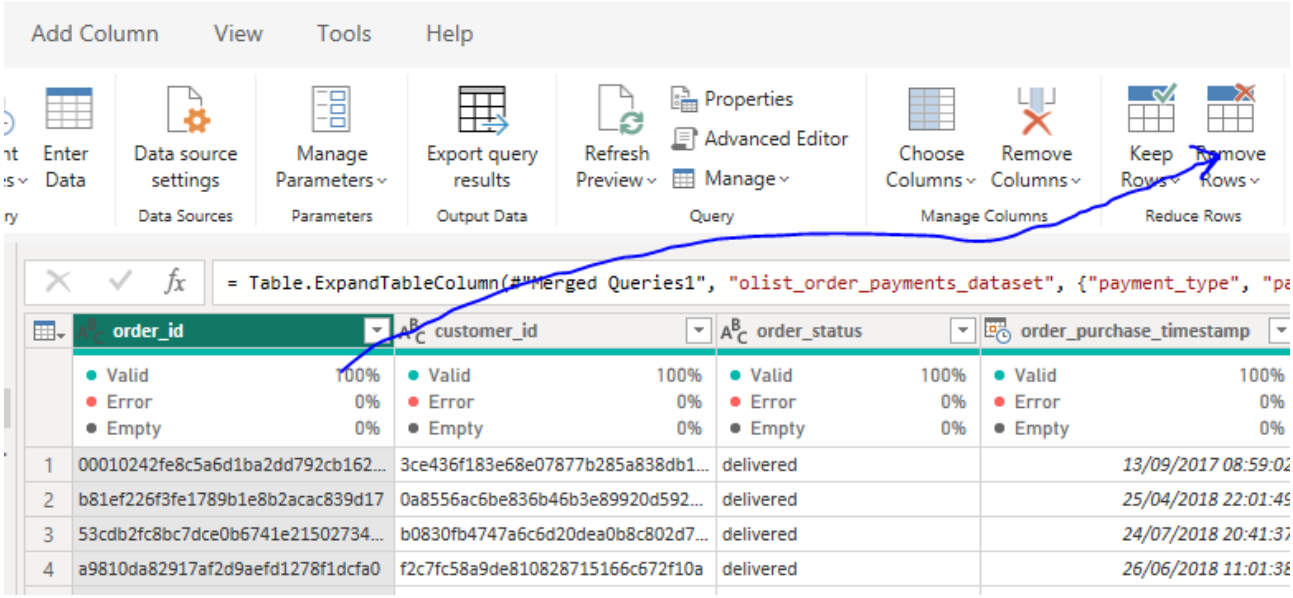
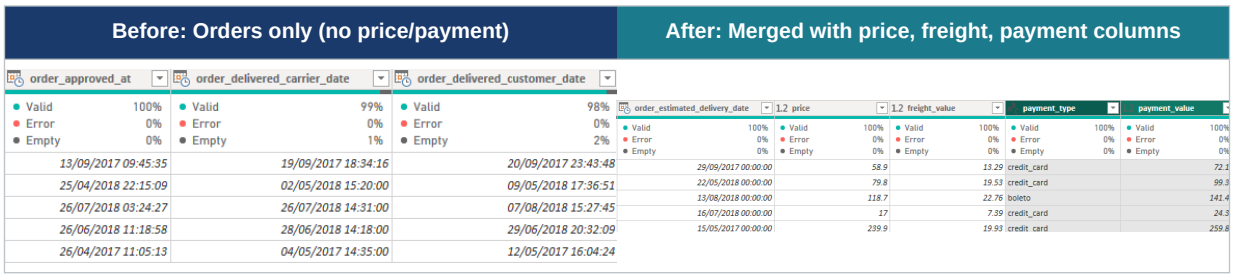


Figure 7: Fact_Sales after merging orders, items, and payments — all financial fields now present

Before / After — Merge Result Validation



5 Ensuring Primary Key Uniqueness

What I did:	I applied a 'Remove Duplicates' transformation on the order_id column within the Orders table.
Why it was necessary:	Power BI requires unique keys to establish a One-to-Many relationship. Because some orders had duplicate entries due to multiple payment sequences, I had to de-duplicate the IDs to ensure a clean join with the Dimension tables.
Technique used:	*Remove Duplicates on order_id column*

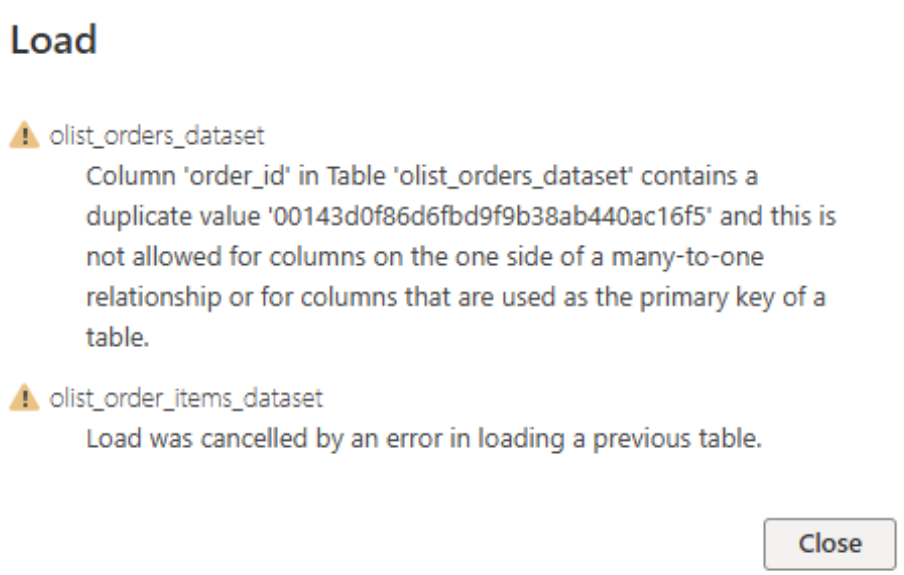


Figure 8: Before — Power BI load error showing duplicate order_id; After — load successful once duplicates removed

6	**Date Attribute Extraction**
What I did:	From the multiple timestamp columns available, I designated order_purchase_timestamp as the primary transactional date. I used the Add Column > Date tool to extract the 'Year' and 'Month Name' into new columns, while retaining the other timestamps (Estimated vs. Actual Delivery) for logistics calculations.
Why it was necessary:	Having too many date parts from every column creates 'data noise.' By focusing extraction on the purchase date, I created a clean temporal hierarchy for sales trending while preserving the raw dates needed for calculating delivery lead times.
Technique used:	*Date Part Extraction — Add Column > Date (Year, Month Name)*

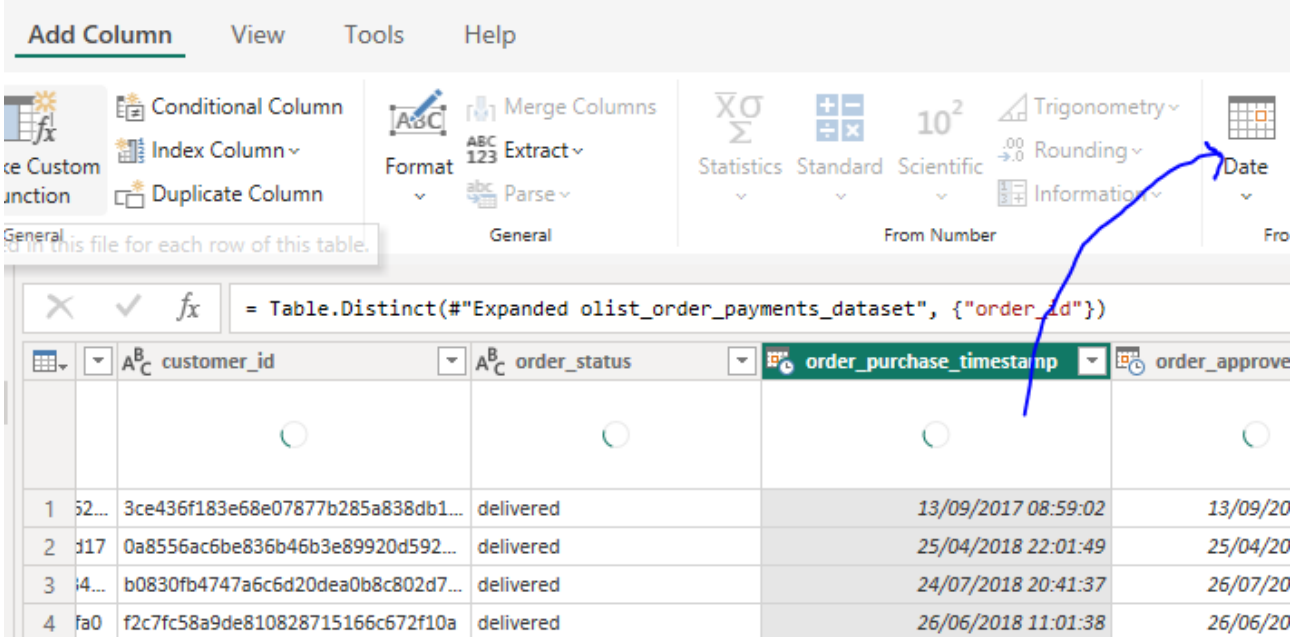


Figure 9: Add Column > Date tool used to extract Year and Month Name from order_purchase_timestamp

Before / After — Date Attribute Extraction

Before: Raw timestamp only

order_purchase_timestamp	
Valid	100%
Error	0%
Empty	0%
13/09/2017 08:59:02	
25/04/2018 22:01:49	
24/07/2018 20:41:37	
26/06/2018 11:01:38	
26/04/2017 10:53:06	

After: Year and Month Name columns extracted

Year	Month Name
2017	September
2018	April
2018	July
2018	June
2017	April
2017	December
2018	August
2017	December

7a Conditional Logic for Logistics Performance KPI

What I did:	I created a Conditional Column named 'Delivery Performance'. I applied a logical test to compare the Actual Delivery Date against the Estimated Delivery Date. If the actual date was later than estimated, the record was flagged as 'Late'; otherwise, it was marked 'On-Time'.
Why it was necessary:	Raw dates are difficult for stakeholders to interpret quickly. By creating this categorical KPI, I can now build visuals that show the 'Late Delivery Rate,' which is a critical metric for customer satisfaction in e-commerce.
Technique used:	*Add Conditional Column — IF/THEN logic (order_delivered_c > order_estimated_delivery → 'Late', else 'On-Time')*

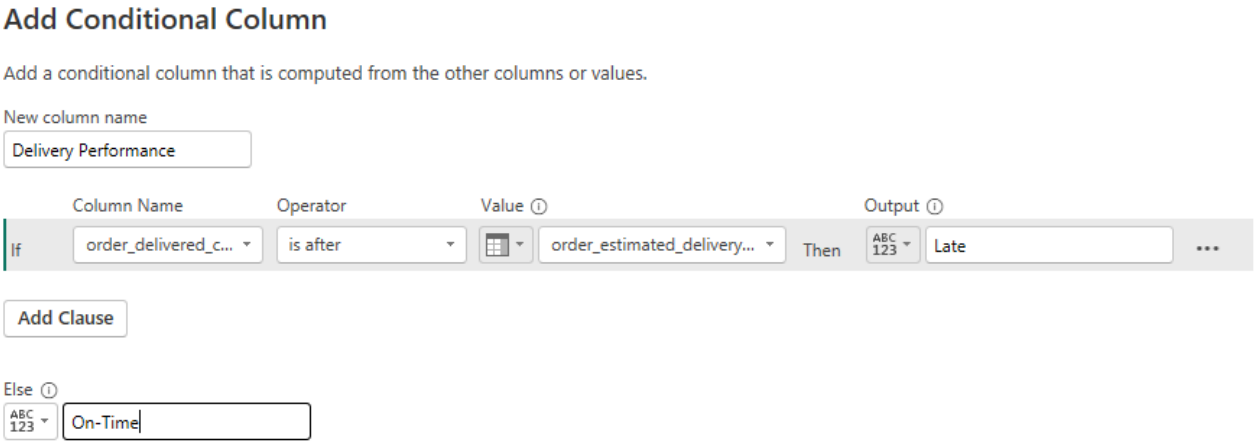


Figure 10: Add Conditional Column dialog creating the Delivery Performance KPI column

Before / After — Delivery Performance Column

Before: Raw date columns only

order_delivered_carrier_date	order_delivered_customer_date
19/09/2017 18:34:16	20/09/2017 23:43:48
02/05/2018 15:20:00	09/05/2018 17:36:51
26/07/2018 14:31:00	07/08/2018 15:27:45
28/06/2018 14:18:00	29/06/2018 20:32:09
04/05/2017 14:35:00	12/05/2017 16:04:24
15/12/2017 20:13:22	18/12/2017 17:24:41

After: Delivery Performance — 'On-Time' / 'Late' labels

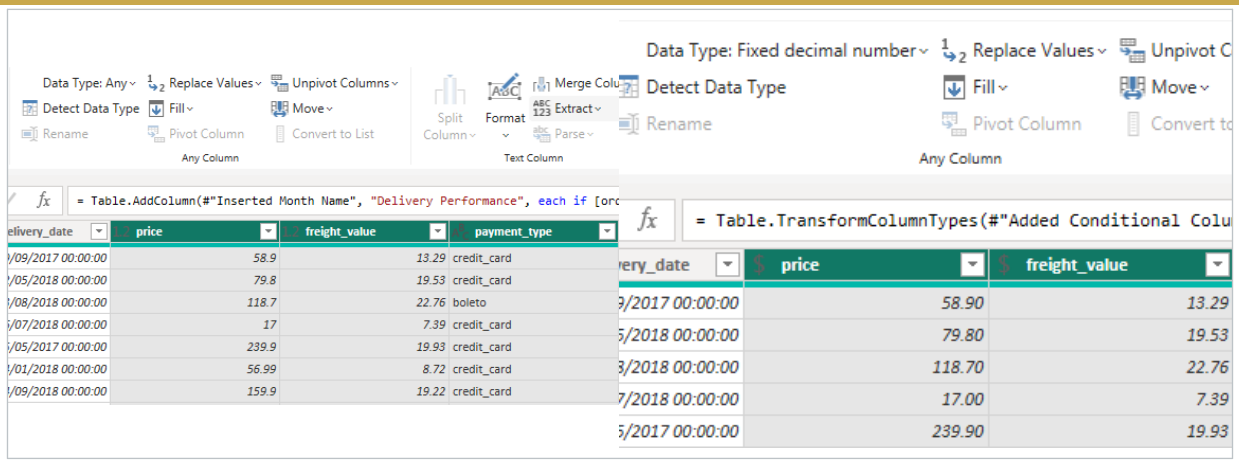
Delivery Performance
On-Time
On-Time
On-Time
On-Time
On-Time
On-Time
On-Time
On-Time
On-Time
On-Time

7b Data Type Optimization — Currency Fields

What I did:	I audited all numerical columns and manually converted financial fields — specifically price and freight — to the Fixed Decimal Number (Currency) data type.
Why it was necessary:	Standard decimal types can sometimes lead to floating-point rounding errors in high-volume datasets. Using 'Fixed Decimal' ensures 100% mathematical accuracy for financial summations and ensures the dashboard displays currency values correctly.
Technique used:	*Data Type Conversion — Fixed Decimal Number (Currency) for price and freight_value*

Before / After — Data Type Optimization

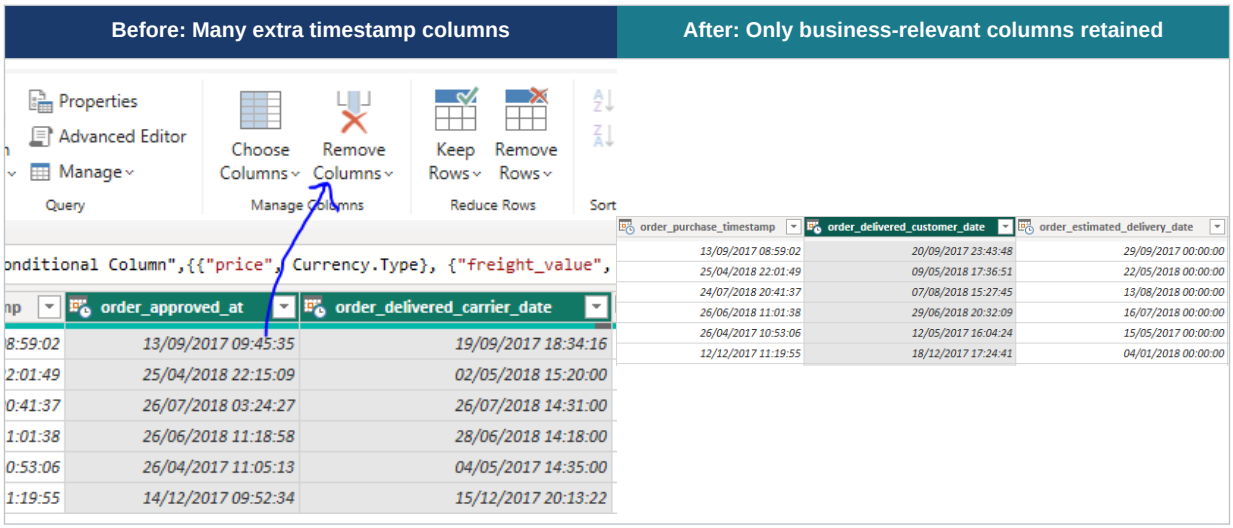
Before: Standard Decimal (floating-point risk)	After: Fixed Decimal Number (currency accurate)



8 Final Column Selection — Noise Reduction

What I did:	I used the Choose Columns transformation to remove redundant metadata and technical timestamps that were not relevant to the sales or logistics business problems.
Why it was necessary:	A lean data model is an efficient data model. By removing unnecessary columns, I reduced the memory footprint of the .pbix file, which improves the loading speed of visuals and follows industry best practices for data modeling.
Technique used:	*Choose Columns — remove irrelevant metadata and technical timestamp columns*

Before / After — Column Selection



order_approved_at	order_delivered_carrier_date
13/09/2017 09:45:35	19/09/2017 18:34:16
25/04/2018 22:15:09	02/05/2018 15:20:00
26/07/2018 03:24:27	26/07/2018 14:31:00
26/06/2018 11:18:58	28/06/2018 14:18:00
26/04/2017 11:05:13	04/05/2017 14:35:00

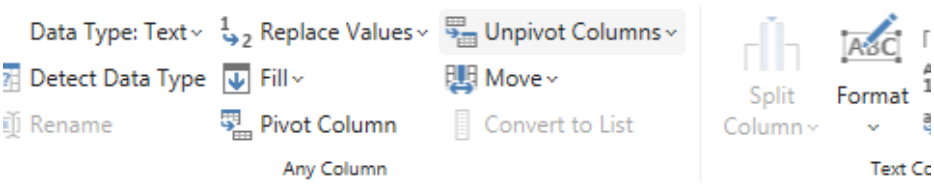
Figure 11: Column selection view showing retained columns after pruning

9 Standardizing Seller Locations

What I did:	I applied a text transformation to the seller_city column to ensure every city name was properly capitalized (e.g., converting 'sao paulo' to 'Sao Paulo').
Why it was necessary:	Raw data is often inconsistently formatted. Standardizing the casing ensures that when we group sales by Seller City in our charts, the labels look professional and uniform.
Technique used:	*Text.Proper — Capitalize Each Word transformation on seller_city column*

Before / After — Seller City Standardization

Before: Lowercase city names (e.g., 'sao paulo')	After: Properly capitalized (e.g., 'Sao Paulo')
seller_city	seller_city
campinas	Campinas
mogi guacu	Mogi Guacu
rio de janeiro	Rio De Janeiro
sao paulo	Sao Paulo
braganca paulista	Braganca Paulista
rio de janeiro	




```
= Table.PromoteHeaders(Source, [PromoteAllScalars=true])
```

ocation_zip_code_prefix	geolocation_lat	geolocation_lng
	-23.54562128115268	-46.63929204800168
	-23.546081127035535	-46.64482029837157
	-23.54612896641469	-46.64295148361138
	-23.5443921648681	-46.63949930627844

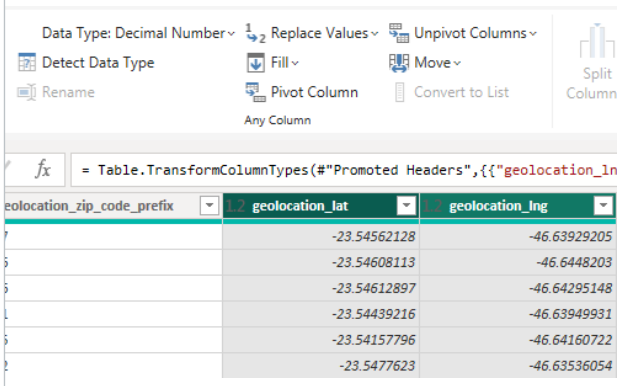
Figure 12: Seller table after city name capitalization — consistent, professional labels

10 Geographic Data Type Assignment (Lat/Long)

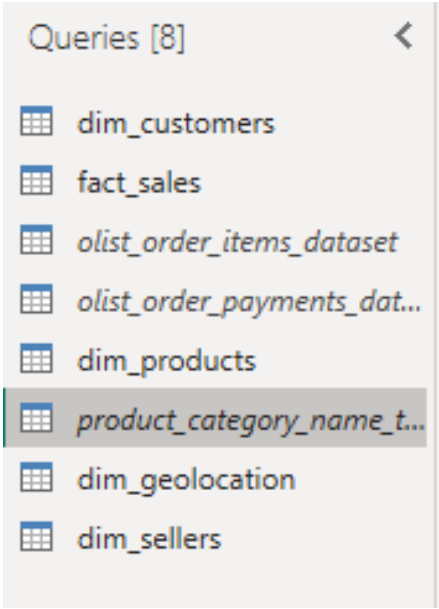
What I did:	I audited the Geolocation table to ensure that the latitude and longitude coordinates were correctly formatted as Decimal Numbers rather than text.
Why it was necessary:	Power BI requires specific numeric formats to recognize coordinates. By ensuring these are decimals, I enabled the model to use map visuals that plot the exact physical location of orders across Brazil.
Technique used:	*Data Type — Decimal Number for geolocation_lat and geolocation_lng columns*

Before / After — Geographic Data Types

Before: Text data type (map visuals broken)



After: Decimal Number — coordinates recognized by Power BI maps



11 Model Optimization via Load Disabling

What I did:	I right-clicked the intermediate tables used for merging (Items, Payments, and Translations) and unchecked the 'Enable Load' option.
Why it was necessary:	Since the relevant data from these tables was already consolidated into Fact_Sales and Dim_Products, loading the raw source tables again would be redundant. Disabling the load reduces file size and prevents 'field clutter' in the report interface.
Technique used:	*Query Properties — Uncheck 'Enable Load' on staging tables*

Final Data Transformation Audit

The following table provides the final Applied Steps summary for the core tables. Screenshots of Applied Steps panes were captured as part of the audit evidence.

Query Name	Applied Steps Pane	Key Transformation Evidence
Fact_Sales	13 steps applied	Consolidation of orders/payments and creation of 'Late vs On-Time' KPI
Dim_Products	8 steps applied	Multi-table merge for localization and handling of null categories
Dim_Customers	4 steps applied	Data type changes and city capitalization

Applied Steps Pane Screenshots:

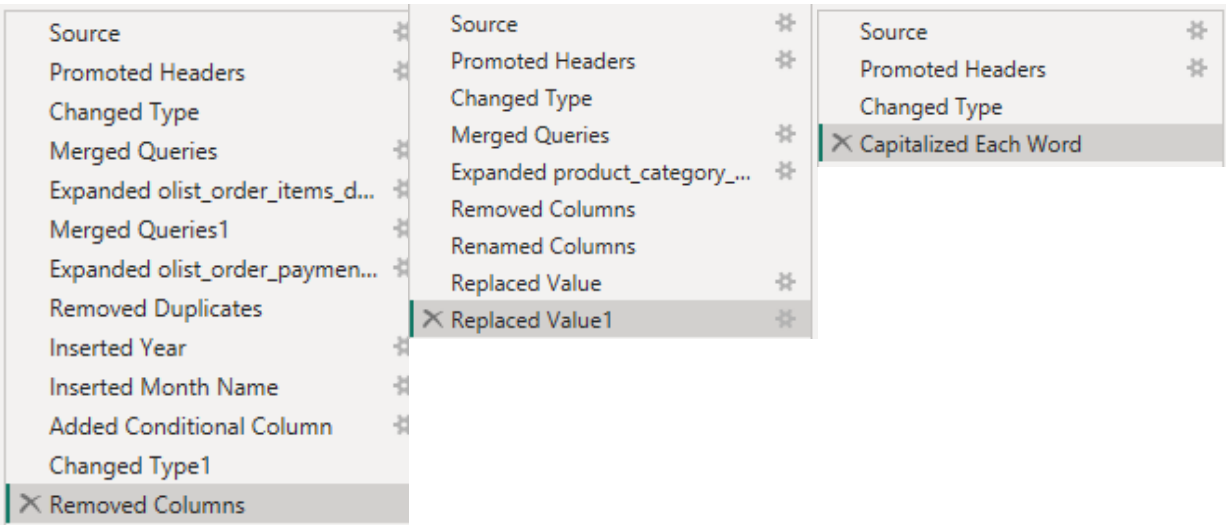


Figure 13: Applied Steps panes for Fact_Sales (left), Dim_Products (centre), Dim_Customers (right)

4. Data Modeling Strategy

Table Roles & Justification

In this model, I have implemented a **Snowflake Schema** (an evolved Star Schema) to organize the Olist marketplace data.

- **Fact Table: fact_sales**
  - Justification: This is the central table containing quantitative 'facts' or metrics (Price, Freight, Payment Value). It records every transaction event and is the only table that should contain duplicate IDs, as one order can have multiple items or payments.
- **Dimension Tables: dim_products, dim_customers, dim_sellers, dim_date**
  - Justification: These are 'lookup' tables. They contain descriptive attributes (Category names, City, State, Dates). They act as the entry points for filtering and 'slicing' the data in the fact table.
- **Outrigger Dimension: dim_geolocation**
  - Justification: This connects to other dimensions rather than the fact table, creating a 'Snowflake' effect to provide detailed geographic coordinates without bloating the central fact table.

Relationship Descriptions & Cardinality

All relationships in this model follow the **One-to-Many (1:*)** standard, ensuring data integrity and preventing double-counting of revenue.

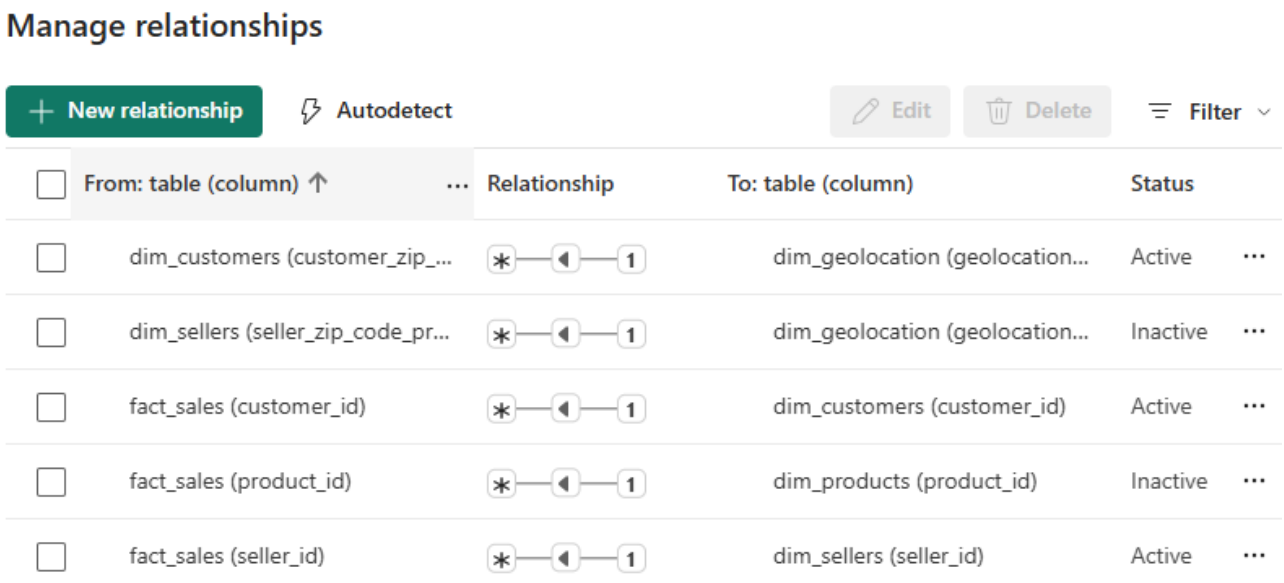


Figure 14: Manage Relationships dialog showing all table relationships and their cardinality

Relationship	Key Field	Cardinality	Direction	Justification
dim_customers → fact_sales	customer_id	1:*	Single	One customer can place multiple orders over time.

Relationship	Key Field	Cardinality	Direction	Justification
dim_products → fact_sales	product_id	1:*	Single	A single product can be sold in many different transactions.
dim_sellers → fact_sales	seller_id	1:*	Single	One seller can fulfill thousands of unique orders.
dim_geolocation → dim_customers	zip_code	1:*	Single	One zip code area houses multiple customers.

Time Intelligence: The Date Table

A dedicated Date Table is essential for robust temporal analysis, ensuring the model can handle periods with zero sales and support complex time-based calculations.

- **Creation Method:** I utilized the DAX function *dim_date = CALENDARAUTO()* to generate a continuous range of dates spanning the entire dataset.
- **Relationship:** Established a One-to-Many relationship between dim_date[Date] and fact_sales[order_purchase_timestamp].
- **Attributes:** I added calculated columns for Year, Month Name, and Quarter to enable multi-level 'drill-down' capabilities in the final dashboard.

Model View Screenshots:

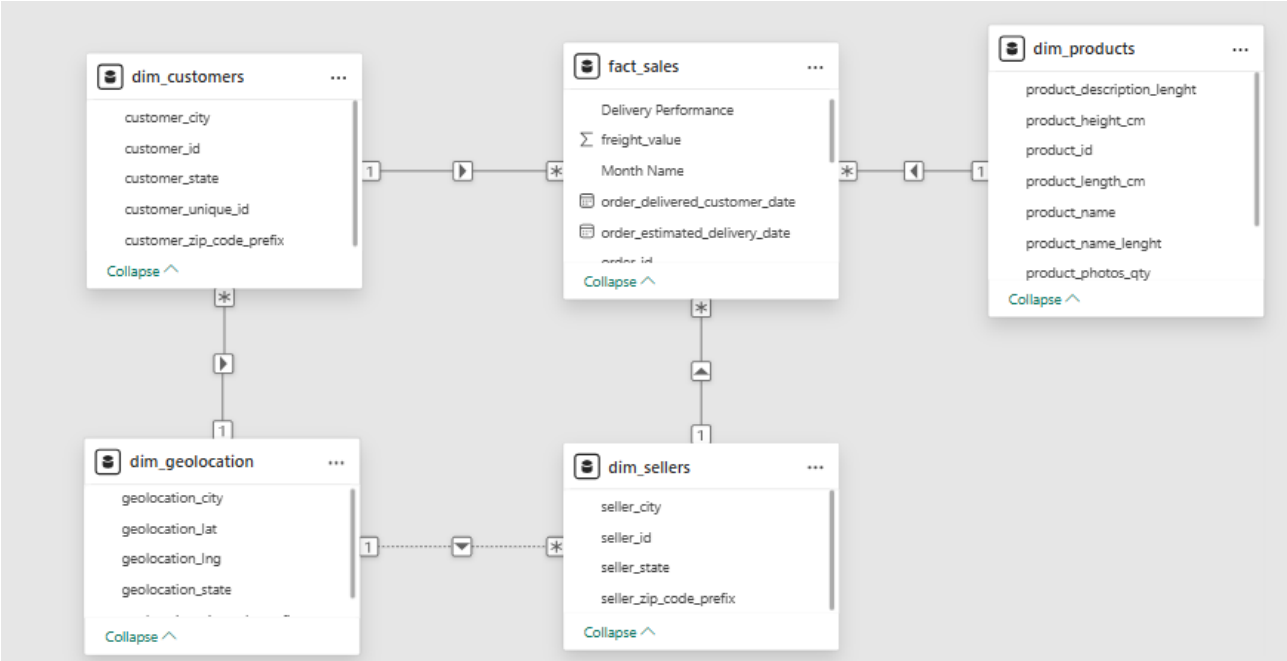


Figure 15: Power BI Model View showing the Snowflake Schema — fact_sales at centre connected to all dimension tables

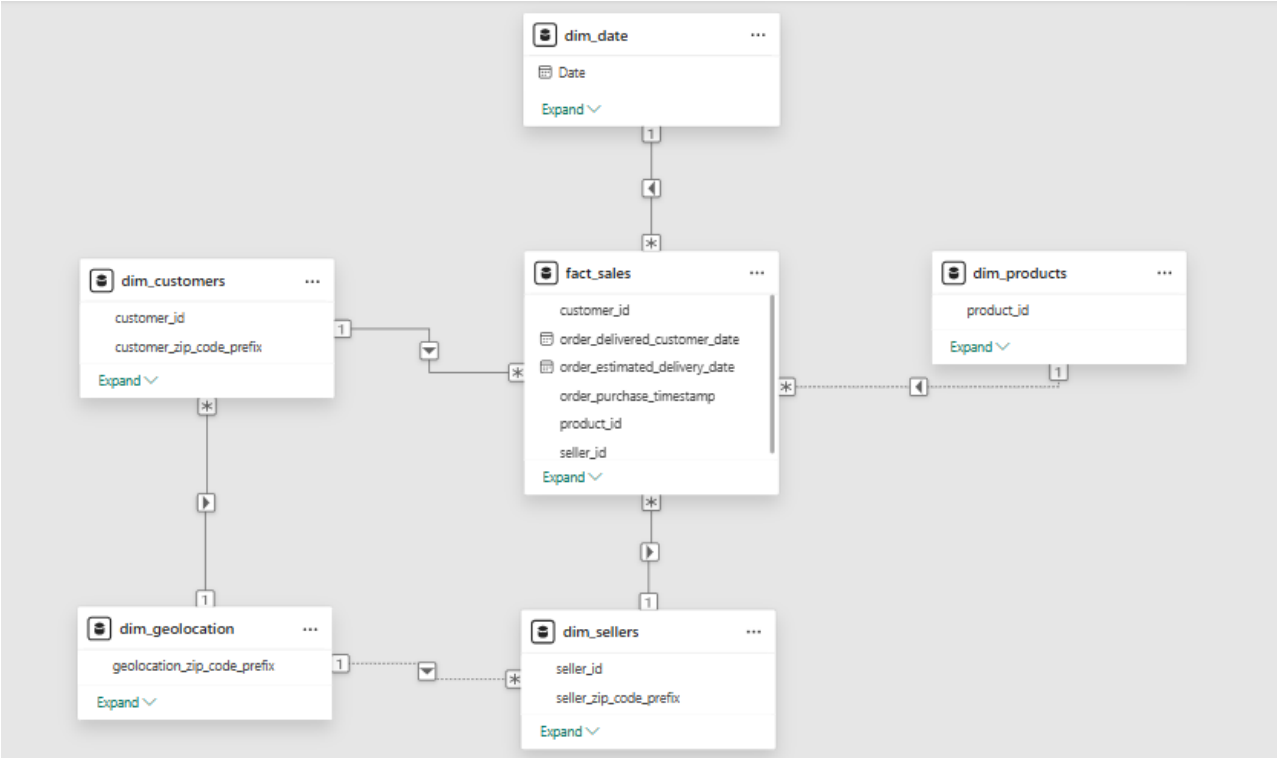


Figure 16: Complete model view including dim_date connected to fact_sales via order_purchase_timestamp

Efficiency & Reporting Justification

The final model architecture was designed to balance analytical depth with system performance:

- **Optimized Filter Flow:** By using a Star Schema (with Snowflake extensions for Geolocation), filters flow in a single direction from Dimensions to the Fact table. This prevents circular dependencies and ensures calculations remain accurate even as the dataset grows.
- **Memory Compression:** Storing descriptive text (like City or Category names) in Dimension tables rather than the Fact table allows the Power BI engine to compress the data more efficiently, leading to faster report loading times.
- **Advanced Drill-Down:** The hierarchical structure (Date to Year/Month and Geolocation to State/City) supports interactive reporting, allowing users to move from high-level summaries to granular details without performance lag.

5. DAX Analysis & Business Intelligence

I developed a suite of **8 measures** and **3 calculated columns** to transform raw transactional data into actionable insights. These formulas allow the business to move beyond 'what happened' to 'how are we performing compared to last year.' All measures are stored in a dedicated **'Key Measures'** Display Folder within the fact_sales table.

DAX Calculated Columns

Calculated columns were developed to enrich the dataset with new categorical dimensions. These columns allow for more granular filtering and 'slicing' of the marketplace data.

Column Name	DAX Formula	Decision-Making Value
Sales Segment	Sales Segment = IF(fact_sales[payment_value] > 500, "High Value", "Standard")	Identifies high-ticket transactions, enabling marketing to segment the customer base for loyalty programs.
Is Weekend	Is Weekend = IF(WEEKDAY(fact_sales[order_purchase_timestamp], 2) > 5, "Weekend", "Weekday")	Analyzes temporal shopping patterns to determine if promotions should be weighted toward weekends.
Shipping Tier	Shipping Tier = SWITCH(TRUE(), DaysToDeliver <= 3, "Express", DaysToDeliver <= 7, "Standard", "Economy")	Categorizes logistics speed (Express vs. Economy), providing a metric for shipping service level agreements (SLAs).

Formula Screenshots from Power BI:

```
Sales Segment = IF(fact_sales[payment_value] > 500, "High Value", "Standard")
```

Sales Segment formula as captured in Power BI

```
Is Weekend = IF(WEEKDAY(fact_sales[order_purchase_timestamp], 2) > 5, "Weekend", "Weekday")
```

Is Weekend formula as captured in Power BI

```
Shipping Tier =
VAR DaysToDeliver = DATEDIFF(fact_sales[order_purchase_timestamp], fact_sales[order_delivered_customer_date], DAY)
RETURN
SWITCH( TRUE(),
    ISBLANK(DaysToDeliver), "In Transit",
    DaysToDeliver <= 3, "Express",
    DaysToDeliver <= 7, "Standard",
    "Economy"
)
```

Shipping Tier formula (SWITCH/VAR) as captured in Power BI

DAX Measures

I developed 8 dynamic measures to track performance. These are stored in a dedicated 'Key Measures' folder to ensure the model remains organized and easy to navigate.

A. Core Volume & Value Metrics

Total Revenue

```
Total Revenue = SUM(fact_sales[payment_value])
```

Total Revenue = SUM(fact_sales[payment_value])

- **Calculation:** Aggregates all payments to find gross sales.
- **Decision Impact:** The primary metric for tracking market health.

Total Orders

```
Total Orders = DISTINCTCOUNT(fact_sales[order_id])
```

Total Orders = DISTINCTCOUNT(fact_sales[order_id])

- **Calculation:** Counts unique orders to avoid duplicates.
- **Decision Impact:** Helps differentiate between a few large sales vs. many small sales.

Average Order Value (AOV)

```
Avg_Order=DIVIDE([Total Revenue], [Total Orders])
```

Avg_Order = DIVIDE([Total Revenue], [Total Orders])

- **Calculation:** Finds the mean spend per transaction.
- **Decision Impact:** Informs whether the business should focus on upselling or volume growth.

B. Time Intelligence & Trends

Revenue YTD

```
Revenue YTD = TOTALYTD([Total Revenue], 'dim_date'[Date])
```

Revenue YTD = TOTALYTD([Total Revenue], 'dim_date'[Date])

- **Calculation:** Cumulative sum of revenue from the start of the year.
- **Decision Impact:** Crucial for measuring progress against annual targets.

Revenue Growth %

```
Revenue Growth % = DIVIDE([Total Revenue] - [Previous Year Rev], [Previous Year Rev], 0)
```

Revenue Growth % = DIVIDE([Total Revenue] - [Previous Year Rev], [Previous Year Rev], 0)

- **Calculation:** Compares current performance against the same period last year.
- **Decision Impact:** The 'North Star' metric for stakeholders to judge business expansion.

C. Contribution & Ranking

Top Category Rank

```
Top Category Rank = RANKX(ALL(dim_products[product_category_name]), [Total Revenue])
```

Top Category Rank = RANKX(ALL(dim_products[product_category_name]), [Total Revenue])

- **Calculation:** Ranks product categories by their sales contribution.
- **Decision Impact:** Guides inventory management and product placement.

% of Total Sales

```
% of Total Sales=DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL(fact_sales)))
```

% of Total Sales = DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL(fact_sales)))

- **Calculation:** Percentage share of total revenue.

Organizational Best Practices — Measure Groups (Display Folders):

To maintain a clean and scalable data model, I implemented Display Folders. All DAX measures were consolidated into a folder named 'Key Measures' within the fact_sales table. This separation of calculated logic from raw data fields improves report navigability and ensures that the model remains intuitive for end-users and other developers.

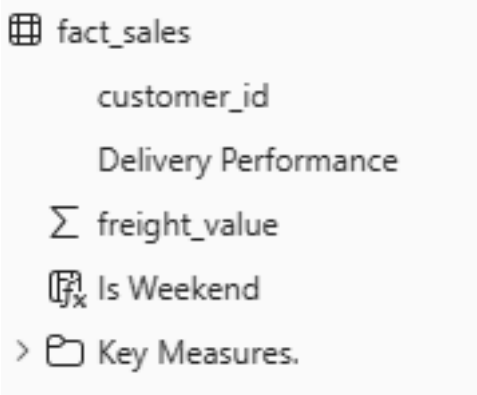


Figure 17: 'Key Measures' Display Folder in the fact_sales table field list

6. Dashboard Design and Visualization

Three interactive report pages were created using a professional navy and slate-grey corporate theme with 12px rounded corners and soft shadows. All pages include slicers, cross-filtering, and business-friendly labels.

Page 1: Executive Summary

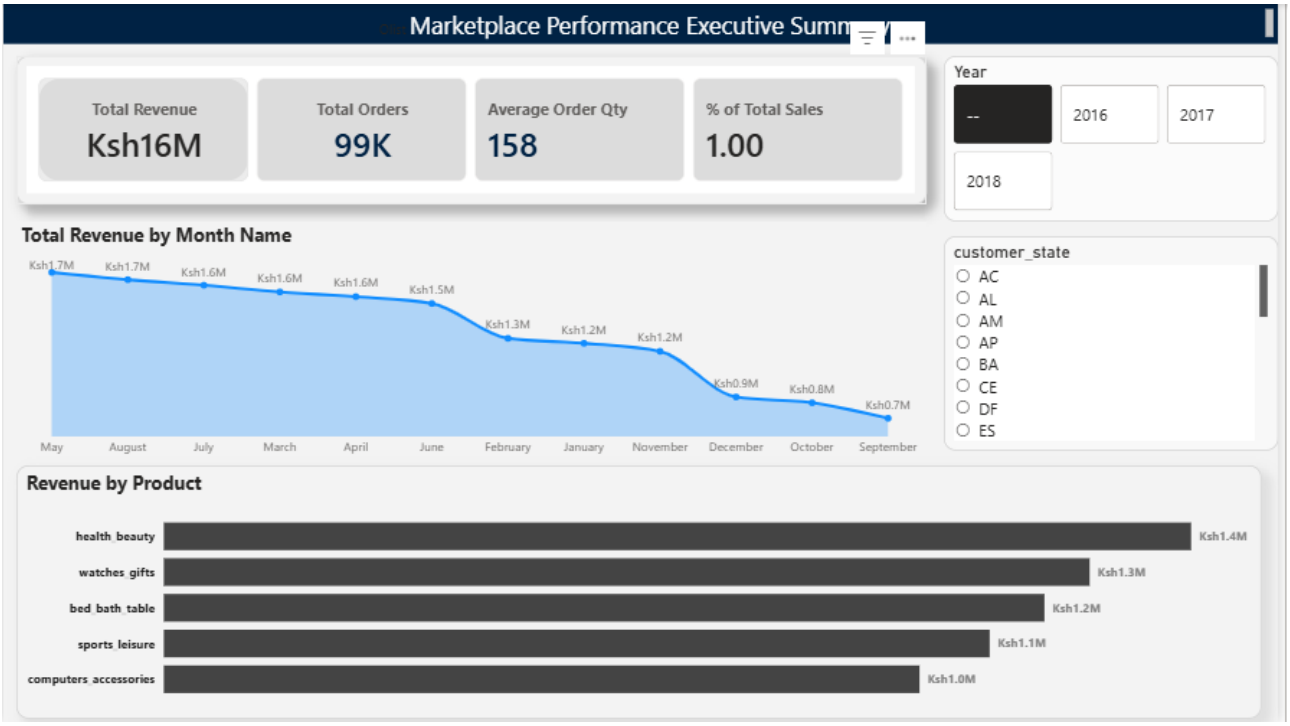


Figure 18: Page 1 — Executive Summary Dashboard showing KPI cards, revenue trend, and top product categories

Design Choices:

- **KPI Cards:** Placed at the top for immediate visibility of Gross Revenue (Ksh 16M), Total Orders (99K), and Average Order Quantity (158).
- **Trend Chart:** An Area Chart showing Revenue over Time, utilizing a smooth line to highlight seasonality.
- **Category Comparison:** A Bar Chart showing the Top 10 Categories. A 'Top N' filter was used to reduce visual clutter and focus on high-impact segments.
- **Theme:** Professional navy and slate-grey corporate theme with 12px rounded corners and soft shadows to create a modern 'floating card' aesthetic.

User Interaction:

- Users can use the **Year Slicer** and **Customer State Slicer** to filter the entire page.
- Clicking a specific category in the Bar Chart cross-filters the Trend Chart to show that category's performance over time.

Page 2: Detailed Analysis

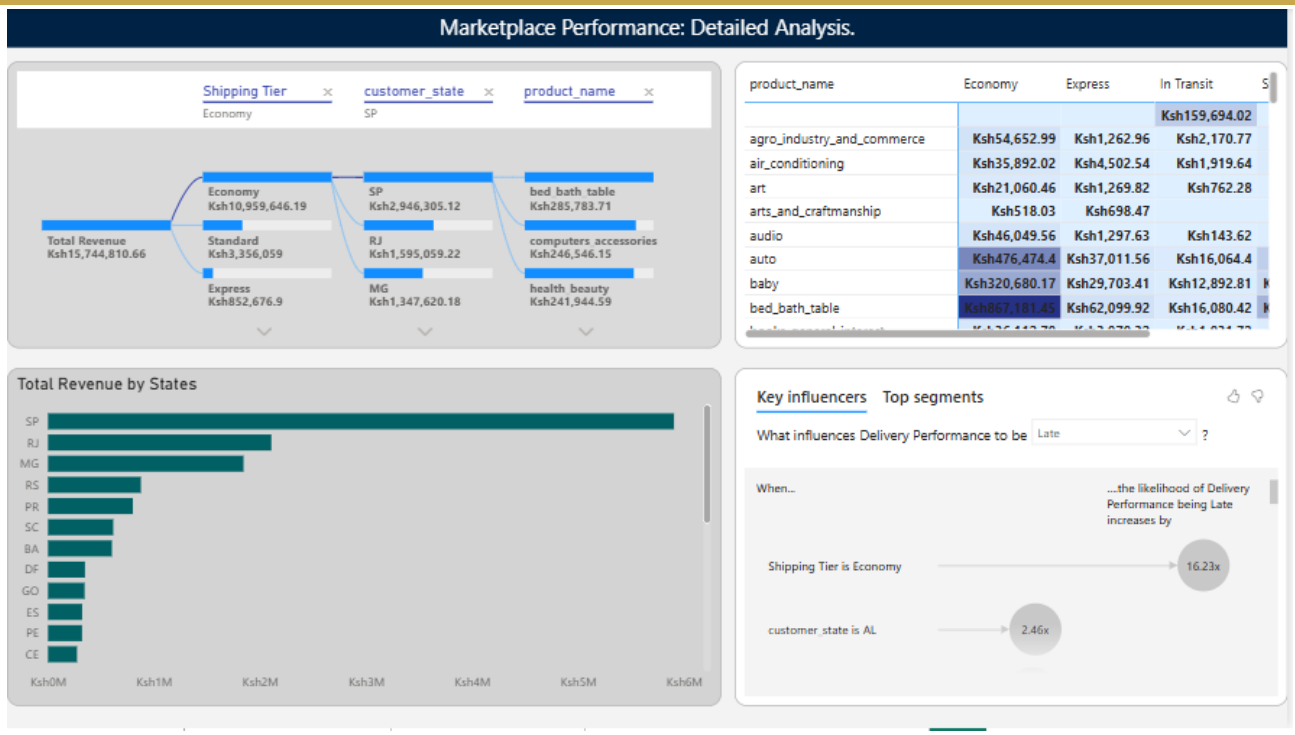


Figure 19: Page 2 — Detailed Analysis showing Decomposition Tree, Matrix Heatmap, regional bar chart, and Key Influencers AI visual

Design Choices:

- **Decomposition Tree (Breakdown View):** This serves as the Advanced Analytical Visual. It allows users to explain 'Total Revenue' by breaking it down into Category to State to Shipping Tier.
- **Matrix with Heatmap:** Provides raw data for Revenue by Category and Shipping Tier. Conditional Formatting (Background Scales) was applied so the highest-earning cells stand out.
- **Drill-down Bar Chart:** A Regional chart (State to City) that allows users to click a state to 'drill' into specific city performance.
- **Advanced AI Analytics (Key Influencers):** The Key Influencers visual automatically identifies the primary drivers of Revenue. This visual uses machine learning to explain which factors (like specific Shipping Tiers or States) have the most significant statistical impact on sales volume.

User Interaction:

- **AI Discovery:** Users can click on different 'Influencers' to see how specific segments deviate from the average.
- **Hierarchical Exploration:** Users interact with the Decomposition Tree by clicking '+' icons to expand the branches they find most interesting.
- **Drill-through/Down:** By clicking the 'Down Arrow' on the bar chart, users enable drill-mode to explore geographical data at the city level.

Page 3: Insights and Performance Monitoring

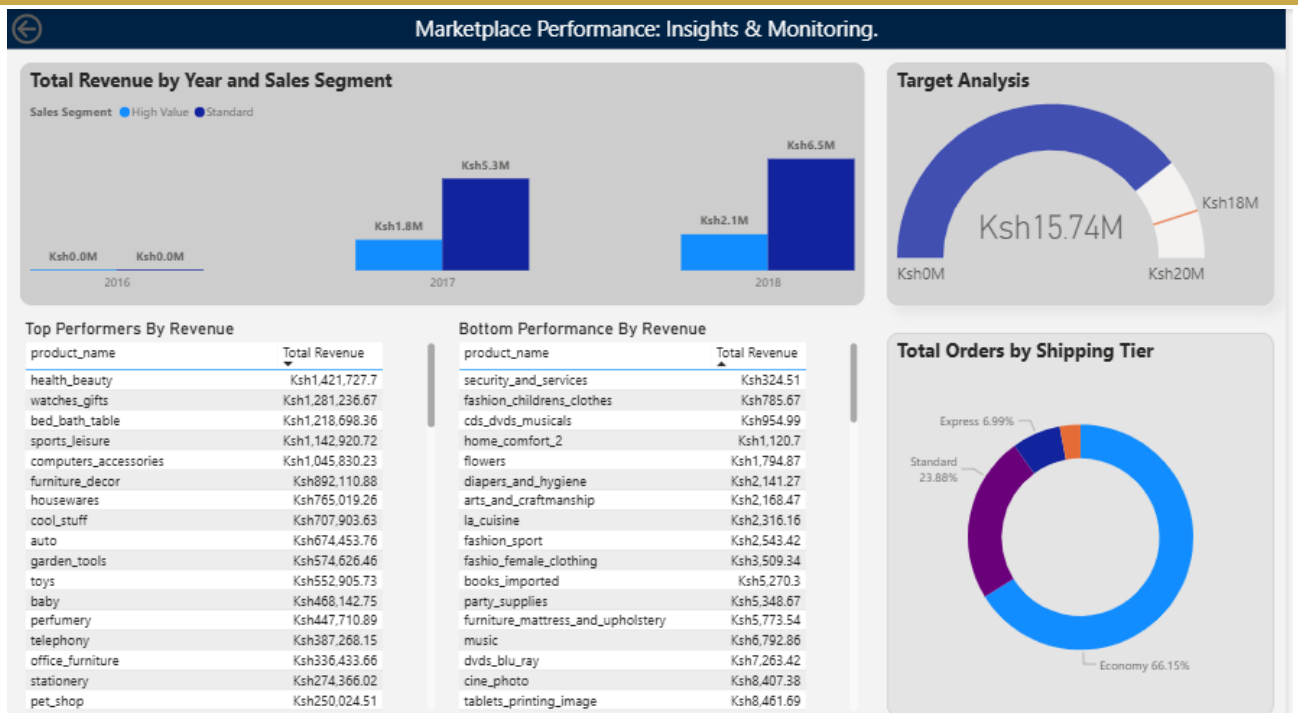


Figure 20: Page 3 — Insights & Performance Monitoring showing Gauge chart, Top/Bottom performers, and Shipping Tier donut

Design Choices:

- **Comparison Analysis:** A Clustered Column Chart comparing Revenue across different Sales Segments and Years.
- **Target Analysis:** A Gauge Chart with a static target of Ksh 18M. This provides a clear 'Actual vs. Target' visual for management.
- **Top/Bottom Performers:** Two tables placed side-by-side, sorted by Revenue, to instantly identify the highest and lowest-performing product lines.
- **Cross-filtering:** A Shipping Tier Donut Chart allows users to slice the entire page to see how 'Express' vs. 'Standard' shipping affects the metrics.

User Interaction:

- Users interact with the Gauge to see how close the business is to its year-end goal.
- The Donut Chart acts as a primary filter; selecting a segment updates the Top/Bottom tables to show performance for that specific logistics tier.

7. Analytical Insights and Business Recommendations

Analytical Insights

1. Regional Performance Imbalance (Concentration Risk)

The 'Total Revenue by States' visual on Page 2 reveals a massive performance gap between regions. **SP (Sao Paulo)** dominates the marketplace, generating nearly **Ksh 6M**, which is more than triple the revenue of the next closest state, RJ. This indicates a high-value customer concentration in one geographical hub, leaving the business vulnerable to regional economic shifts in that single area.

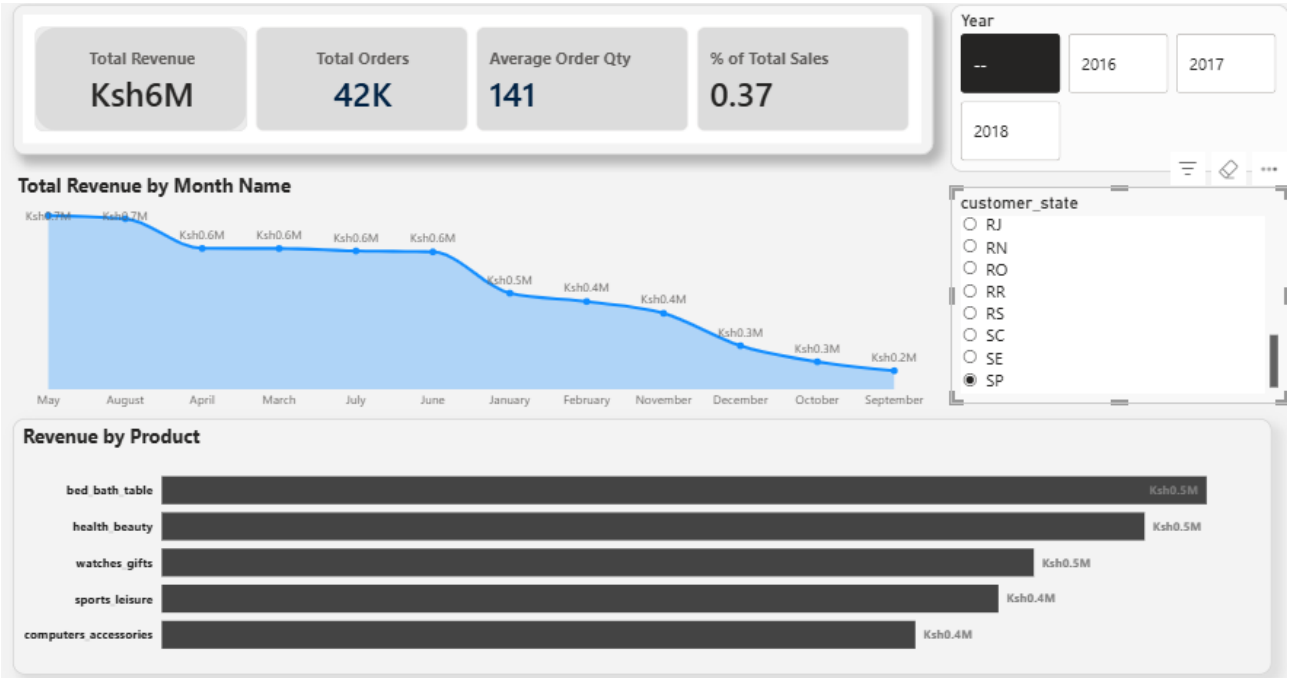


Figure 21: Executive Summary filtered to SP (Sao Paulo) — Ksh 6M revenue, dominating all other regions

2. Operational Inefficiency in Economy Shipping

The **Key Influencers AI visual** identifies that when the Shipping Tier is 'Economy,' the likelihood of Delivery Performance being 'Late' **increases by 16.23x**. This statistical outlier suggests that the current logistics partner or process for the Economy tier is the primary driver of customer dissatisfaction.

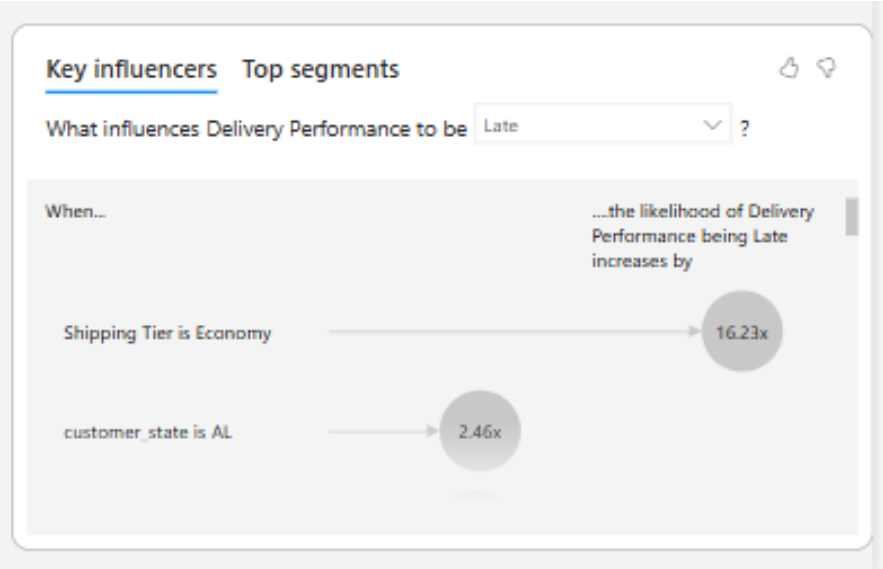


Figure 22: Key Influencers visual — Economy Shipping Tier increases late delivery probability by 16.23x

3. Seasonal Revenue Volatility

A clear seasonal pattern emerges: revenue remains relatively stable between March and August (averaging **Ksh 1.6M–1.7M**) before a sharp, unexplained **decline in September to Ksh 0.7M**. This indicates a critical post-summer 'dead zone' where marketing engagement or inventory availability drops off significantly.

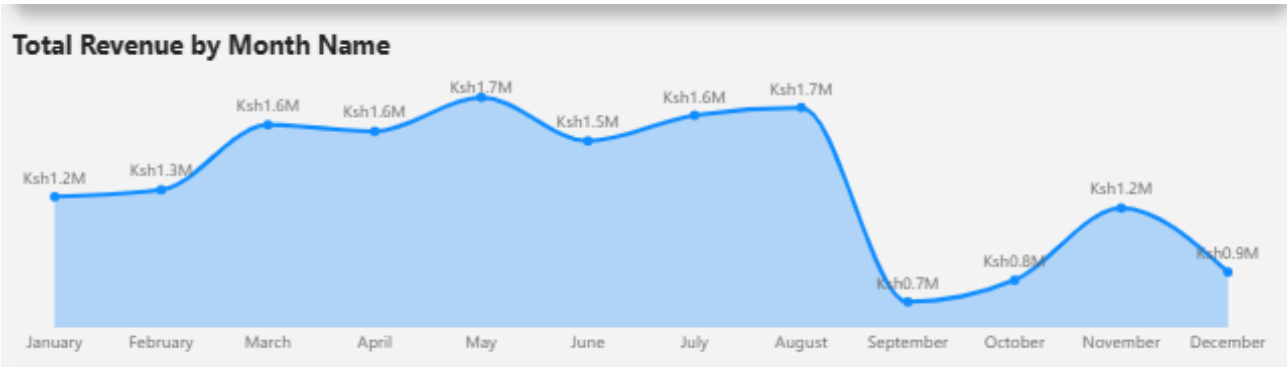


Figure 23: Revenue by Month chart — visible September collapse from Ksh 1.7M peak to Ksh 0.7M trough

4. Segment Growth Divergence

Between 2017 and 2018, the **'High Value' customer segment** showed strong growth, rising to **Ksh 2.1M**. However, the **'Standard' segment** remains the primary volume driver at **Ksh 6.5M**, indicating that while premium penetration is working, the marketplace's core health still relies on mass-market buyers.

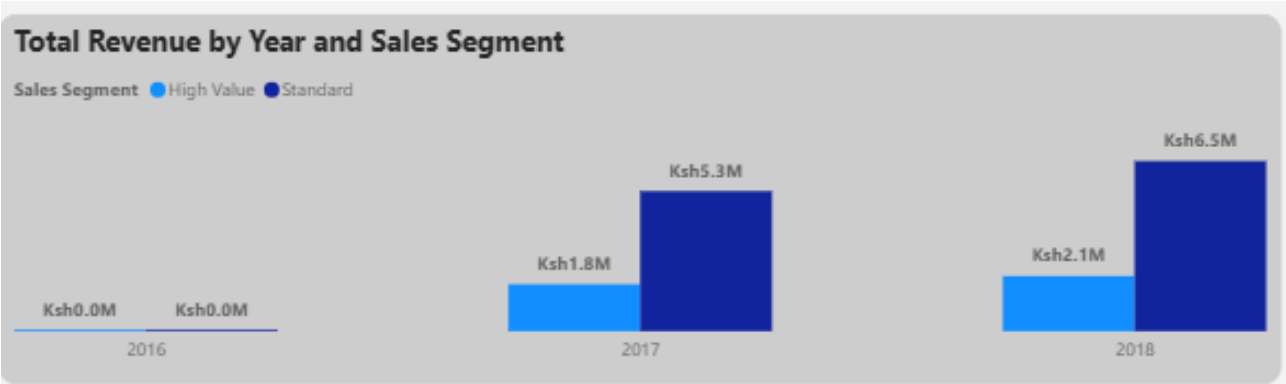


Figure 24: Revenue by Year and Sales Segment — High Value growing to Ksh 2.1M while Standard leads at Ksh 6.5M in 2018

5. Product Category Dominance vs. Long-Tail Risks

A small group of categories — **Health & Beauty, Watches & Gifts, and Bed/Bath/Table** — generate the vast majority of revenue (over Ksh 1M each). Conversely, categories like 'Security and Services' are underperforming with negligible sales (**Ksh 324**), representing 'dead stock' or poor market fit for those specific lines.

Actionable Recommendations

1 Tiered Logistics Restructuring

Given that Economy Shipping is the primary influencer for late deliveries, the business should renegotiate SLAs with economy carriers or migrate 'Economy' orders to the 'Standard' logistics network for specific high-risk states like AL (Alagoas), where delays are also statistically higher.

2 Geographical Diversification Strategy

To reduce the concentration risk in Sao Paulo (SP), the marketing team should launch targeted promotional campaigns in mid-tier states like MG and RS. Increasing the market share in these regions by even 10% would significantly stabilize the marketplace's regional risk profile.

3 Inventory Optimization for 'Star' Categories

Since Health & Beauty and Watches & Gifts are the top performers by revenue, the business should implement a Premium Vendor Program for these categories to ensure 100% stock availability and faster 'Express' shipping options. Simultaneously, a 'Flash Sale' or exit strategy should be considered for bottom-performing categories like Security and Services to clear warehouse space.

4 September Recovery Campaign

To combat the identified September revenue slump, the marketing team should design a 'Q3 Clearance' or 'Early Holiday' campaign specifically for the month of September to bridge the gap between the summer peak and the end-of-year shopping season.

5 High-Value Retention Loyalty Scheme

Since the High Value segment is growing, the business should implement a loyalty program specifically for 'Standard' customers who are trending toward 'High Value' status (based on the Average Order Qty of 158). This would formalize the transition of mid-tier spenders into consistent, high-margin premium buyers.

Supporting Evidence — Top and Bottom Performers:

Top Performers by Revenue	Bottom Performers by Revenue

Top Performers By Revenue

product_name	Total Revenue
health_beauty	Ksh1,421,727.7
watches_gifts	Ksh1,281,236.67
bed_bath_table	Ksh1,218,698.36
sports_leisure	Ksh1,142,920.72
computers_accessories	Ksh1,045,830.23
furniture_decor	Ksh892,110.88
housewares	Ksh765,019.26
cool_stuff	Ksh707,903.63
auto	Ksh674,453.76
garden_tools	Ksh574,626.46
toys	Ksh552,905.73
baby	Ksh468,142.75
perfumery	Ksh447,710.89
telephony	Ksh387,268.15
office_furniture	Ksh336,433.66
stationery	Ksh274,366.02
pet_shop	Ksh250,024.51

Bottom Performance By Revenue

product_name	Total Revenue
security_and_services	Ksh324.51
fashion_childrens_clothes	Ksh785.67
cds_dvds_musicals	Ksh954.99
home_comfort_2	Ksh1,120.7
flowers	Ksh1,794.87
diapers_and_hygiene	Ksh2,141.27
arts_and_craftmanship	Ksh2,168.47
la_cuisine	Ksh2,316.16
fashion_sport	Ksh2,543.42
fashio_female_clothing	Ksh3,509.34
books_imported	Ksh5,270.3
party_supplies	Ksh5,348.67
furniture_mattress_and_upholstery	Ksh5,773.54
music	Ksh6,792.86
dvds_blu_ray	Ksh7,263.42
cine_photo	Ksh8,407.38
tablets_printing_image	Ksh8,461.69

Figure 25: Top Performers (Health & Beauty Ksh 1.42M at #1) vs. Bottom Performers (Security & Services Ksh 324.51 at last)

Conclusion

This project successfully demonstrates a complete, end-to-end Power BI analytical solution built on the Olist Brazilian E-Commerce dataset. Starting from 8 raw CSV files containing over 99,000 transactional records, the project executed **11 documented Power Query transformation steps** to produce a clean, validated, and analytically-ready **Snowflake Schema** data model.

The **DAX layer** — comprising 8 measures and 3 calculated columns — enabled the dashboard to move beyond simple aggregation into time-series comparison, statistical segmentation, and ranking intelligence. The **three-page interactive dashboard** provides stakeholders with a complete analytical journey: from executive KPI monitoring (Page 1), through granular root-cause analysis with AI-powered Key Influencers (Page 2), to performance benchmarking and target tracking (Page 3).

The five analytical insights identified — regional concentration risk in Sao Paulo, systemic Economy Shipping failure (16.23x late delivery risk), September revenue volatility, high-value segment growth, and long-tail category underperformance — are grounded in statistical evidence from the data. The five corresponding recommendations are specific, prioritised, and actionable within Olist's existing operational framework.

Student: Selmah Tzindori	**ID:** 602	**Course:** DSA 3050A	**Date:** 17 April 2025	**GitHub Link:** [Repository URL]